WHY PYBIO-10

From the bench to the code.

Spreadsheets break when the data gets big. In 10 evenings you learn to clean, analyse, visualise and model your own biological data in Python, the language used across modern genomics, bioinformatics and lab analytics.

```python
# Day 7: your first sequence scan
from Bio import SeqIO
from Bio.SeqUtils import gc_fraction
for r in SeqIO.parse("genes.fasta", "fasta"):
    print(r.id, gc_fraction(r.seq))
```

By Day 10 you will be able to

- Write clean, reusable Python in Jupyter and Google Colab
- Turn messy lab and field data into tidy, analysis-ready tables
- Run t-tests, ANOVA and regression, and make publication-ready figures
- Read FASTA and GenBank files, run BLAST, and handle gene-expression data
- Build and evaluate your first machine-learning model for biology

AVYAGRAHA
RESEARCH AND ANALYTICS LLP

AVYAGRAHA RESEARCH AND ANALYTICS LLP PRESENTS

REGISTRATIONS OPEN

PyBio-10

Python for Life Sciences

10-DAY HANDS-ON TRAINING PROGRAMME

DATES
16 – 26 Nov 2026
Sunday 22 Nov holiday

TIMING
6:30 – 8:30 PM IST
2 hours per evening

MODE
Online · Live
Hands-on, no prior coding

FEE (INCL. GST)
₹1,499 / ₹1,999
Students / Professionals

RECORDINGS
3-month access
Revisit every session

CERTIFICATE
E-certificate
Plus capstone projects

LAST DATE TO REGISTER 15 Nov 2026, 5:00 PM IST

`Python` `NumPy` `Pandas` `Matplotlib` `Seaborn` `SciPy` `Biopython` `scikit-learn`

BONUS 20+ AI tools for research

WHO CAN REGISTER

Built for life scientists, not programmers.

- UG, PG and PhD students in the life sciences
- Faculty, scientists and research associates
- Biotechnology, Microbiology, Biochemistry, Genetics, Bioinformatics, Botany, Zoology, Agriculture, Horticulture, Forestry, Veterinary, Fisheries, Food Science, Pharmacy, Medical and allied life sciences
- Pharma, agri-biotech and healthcare professionals
- Anyone moving from Excel or SPSS to code-based analysis

No prior programming experience needed. A laptop with internet access and a Google account (for Colab) is all you need.

10	20 h	11
live evening sessions	of hands-on coding	tools and libraries

Start where you are.
Code your own data.

ENROLL NOW

aviqlabs.com/pybio10

Students
UG / PG / PhD students and research scholars
₹1,499

Professionals
Faculty, scientists, industry and others
₹1,999

All fees inclusive of GST · Limited seats

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3. Note the **UPI transaction ID** after paying
4. Enter it, **upload the payment screenshot** and submit

Questions?

Call or WhatsApp
+91 83105 79116 · +91 94803 63456
Email **avinashstat@aviqlabs.com**

COURSE STRUCTURE

10 days. 10 skills.

Every session is live and hands-on: you write and run the code yourself, on biological datasets.

Module 1 · Python Foundry Days 1–2

D01 Inauguration · Python foundations
Colab notebooks, variables, data types, operators
Mon 16 Nov

D02 Thinking in code
Data structures, conditions, loops, functions
Tue 17 Nov

Module 2 · Scientific Data Toolkit Days 3–5

D03 NumPy for scientific computing
Arrays, broadcasting, vectorised calculations
Wed 18 Nov

D04 Pandas for lab data
Clean, validate, merge and reshape datasets
Thu 19 Nov

D05 Publication-ready visualisation
Matplotlib, Seaborn, multi-panel figures
Fri 20 Nov

Module 3 · Statistics and Sequences Days 6–7

D06 Biostatistics in Python
t-tests, ANOVA, regression, assumptions
Sat 21 Nov

D07 Biopython: sequence analysis
FASTA, GenBank, translation, BLAST
Mon 23 Nov

Module 4 · Omics and ML Days 8–9

D08 Omics data handling
Expression matrices, FDR, volcano plots
Tue 24 Nov

D09 Machine learning for biology
Classification, evaluation, clustering, PCA
Wed 25 Nov

Module 5 · Capstone Day 10

D10 Capstone and valedictory
End-to-end analysis, presentations, e-certificates
Thu 26 Nov

Sunday 22 November — HOLIDAY · NO SESSION — Every session: 6:30 – 8:30 PM IST

TOOLS YOU WILL USE: `Python 3` `Jupyter` `Google Colab` `NumPy` `Pandas` `Matplotlib` `Seaborn` `SciPy` `statsmodels` `Biopython` `scikit-learn`

COORDINATOR AND RESOURCE PERSONS

Learn from practitioners.



PROGRAMME COORDINATOR

Dr. Avinash Gopa

Founder & CEO, Avyagraha Research & Analytics LLP · Former ICMR Project Scientist-II · Ph.D. (Agricultural Statistics), IARI, New Delhi



RESOURCE PERSON

Mr. Chandan G Y

Data Analyst, Bangalore



RESOURCE PERSON

Mr. Rajesh A

CEO & Tech Member, Avinya Techsphere, Bangalore



RESOURCE PERSON

Mr. Shrey Pandya

Research Analyst, Population Research Centre, Department of Statistics, The Maharaja Sayajirao University of Baroda, Vadodara



RESOURCE PERSON

Dr. Sandip Garai

Scientist, ICAR-Indian Institute of Agricultural Biotechnology, Garhkhatanga, Ranchi

KEY TAKEAWAYS

You walk away with

3M — 3-month recordings
Revisit any session at your own pace

.py — Code notebooks
Every day's notebook, ready to reuse on your own research data

Q&A — Live doubt clearing
Direct interaction with the trainers in every session

CP — Capstone projects
End-to-end analyses you can show on your CV

EC — E-certificate
Issued on completion of the programme

DNA — Real biological datasets
Sequence, expression, lab and field data from published studies

AI — Bonus: 20+ AI tools for research
A curated introduction to AI tools that support research work

ABOUT AVYAGRAHA

AVYAGRAHA
RESEARCH AND ANALYTICS LLP

A DPIIT-recognised deep-tech startup, incubated at AIC-UASB. We provide data-driven solutions, training programmes and research collaborations.

- **Training:** statistics, R, Python, AI/ML and Generative AI for institutions and professionals
- **Research consultancy:** study design, data analysis and statistical modelling
- **Partnerships:** MoUs with universities and R&D institutes

Register now

Scan the QR, pay by UPI and upload your payment screenshot.

`aviqlabs.com/pybio10`

Contact us

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avinashstat@aviqlabs.com

www.aviqlabs.com

DETAILED TRAINING PROGRAMME CURRICULUM

PyBio-10: Python for Life Sciences
10-Day Hands-On Training Programme

16 Nov 2026 – 26 Nov 2026 · 6:30 PM to 8:30 PM IST · Online (live) · Sunday holiday
Organized by Avyagraha Research and Analytics LLP, Ramasagara, Ballari, Karnataka
www.aviqlabs.com · avinashstat@aviqlabs.com · +91 83105 79116 · +91 94803 63456

10 live sessions | 20 hours hands-on | 3-month recordings | Capstone projects | E-certificate

Date	Topics	Resource person
Module 1 · Python Foundry (Days 1–2)		
16/11/2026 Monday	D01 **Inauguration and Python foundations** Colab setup, notebooks, variables, data types and operators. Practical: a documented notebook of sample data.	**Dr. Avinash Gopa** Founder & CEO, Avyagraha & **Mr. Chandan G Y** Data Analyst, Bangalore
17/11/2026 Tuesday	D02 **Thinking in code: data structures and logic** Lists, dictionaries, conditions, loops and functions. Practical: count DNA bases and convert assay units.	**Mr. Rajesh A** CEO & Tech Member, Avinya Techsphere, Bangalore
Module 2 · Scientific Data Toolkit (Days 3–5)		
18/11/2026 Wednesday	D03 **NumPy for scientific computing** Arrays, indexing, broadcasting and vectorised calculations. Practical: process a plate-reader absorbance matrix.	**Mr. Rajesh A** CEO & Tech Member, Avinya Techsphere, Bangalore
19/11/2026 Thursday	D04 **Pandas for biological and laboratory data** Read, clean, validate, merge and reshape lab data. Practical: a tidy lab dataset joined with sample metadata.	**Mr. Chandan G Y** Data Analyst, Bangalore
20/11/2026 Friday	D05 **Publication-ready scientific visualisation** Box, violin and scatter plots, heatmaps and multi-panel figures, exported at high resolution.	**Dr. Avinash Gopa** Founder & CEO, Avyagraha
Module 3 · Statistics and Sequences (Days 6–7)		
21/11/2026 Saturday	D06 **Biostatistics in Python** Descriptive statistics, t-tests, chi-square, ANOVA with Tukey post-hoc, correlation and regression.	**Mr. Shrey Pandya** Research Analyst, Dept. of Statistics, MSU Baroda, Vadodara
22/11/2026 Sunday	HOLIDAY	

Date	Topics	Resource person
23/11/2026 Monday	D07 **Biopython essentials: sequence analysis** DNA, RNA and protein sequences; FASTA and GenBank parsing, GC content, translation and BLAST.	**Dr. Avinash Gopa** Founder & CEO, Avyagraha
Module 4 · Omics and Machine Learning (Days 8–9)		
24/11/2026 Tuesday	D08 **Omics data handling** Expression matrices, normalisation, fold change and FDR, with volcano plots and clustered heatmaps.	**Dr. Avinash Gopa** Founder & CEO, Avyagraha
25/11/2026 Wednesday	D09 **Machine learning for biology** Logistic regression and random forest, model evaluation, clustering and PCA.	**Dr. Sandip Garai** Scientist, ICAR-IIAB, Ranchi
Module 5 · Capstone (Day 10)		
26/11/2026 Thursday	D10 **Capstone, certification and valedictory** An end-to-end analysis, short presentations, feedback and e-certificates.	**Dr. Avinash Gopa** Founder & CEO, Avyagraha

Programme fees

Category	Fee (incl. GST)
Students: UG / PG / PhD and research scholars	**Rs. 1,499 /-**
Professionals: faculty, scientists, industry, others	**Rs. 1,999 /-**

Requirements

- A laptop or desktop with a stable internet connection.
- A Google account, for Google Colab notebooks.
- No software to install: everything runs in the browser.

How to register

1. Scan the QR code or visit **aviqlabs.com/pybio10**.
2. Tap **Pay now** (any UPI app opens with the fee filled in), or pay to UPI ID **9480363456@okbizaxis** (Avyagraha Research and Analytics LLP).
3. Note the **UPI transaction ID** shown after payment.
4. Enter the transaction ID, **upload the payment screenshot** and submit to get your registration ID by email.
5. Joining details will be shared by email and WhatsApp before the programme begins.

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